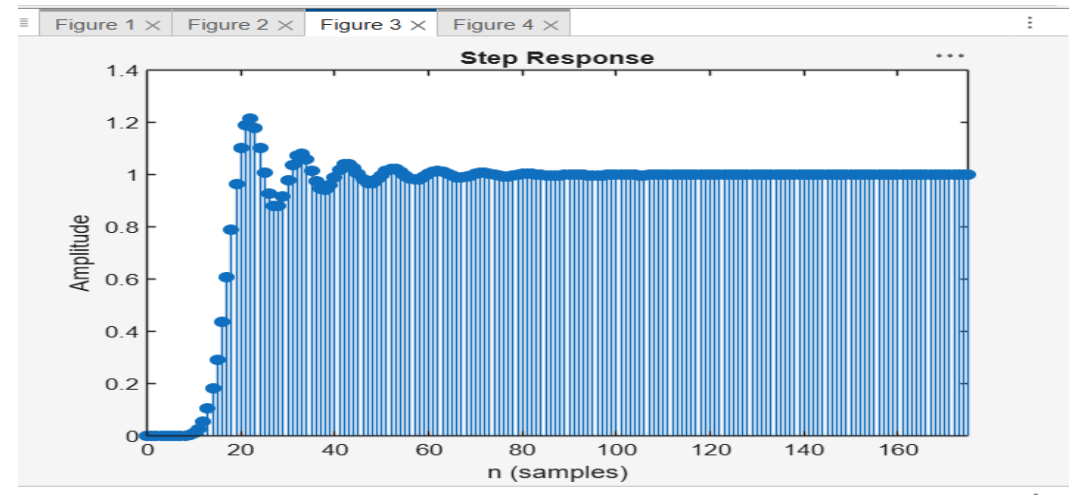
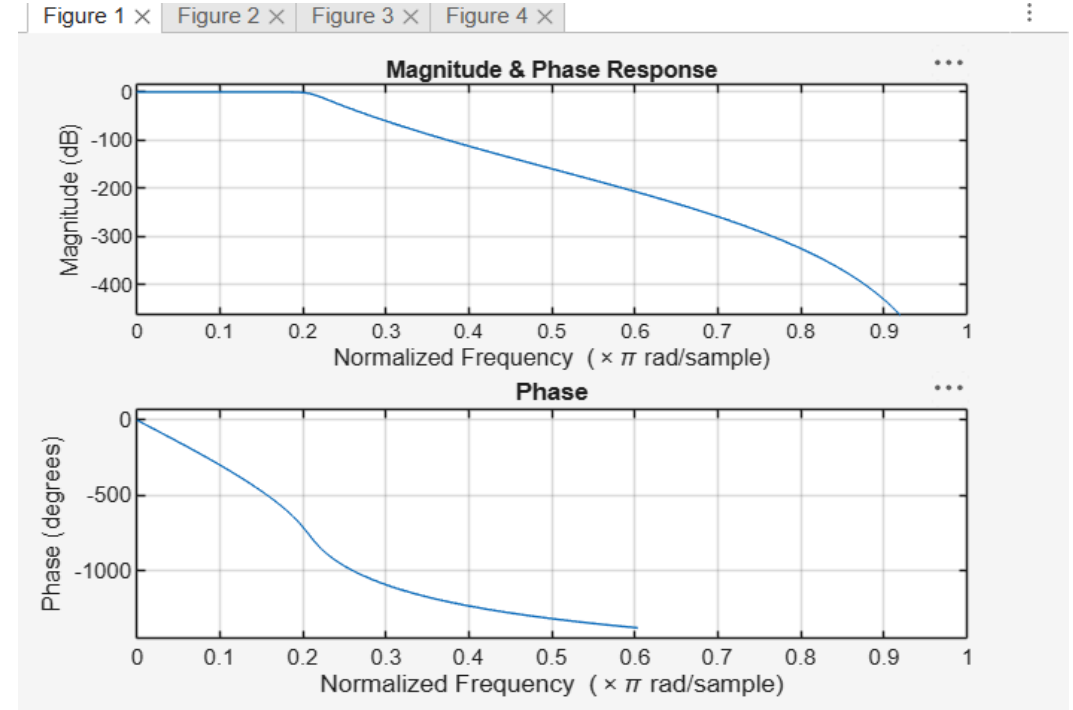
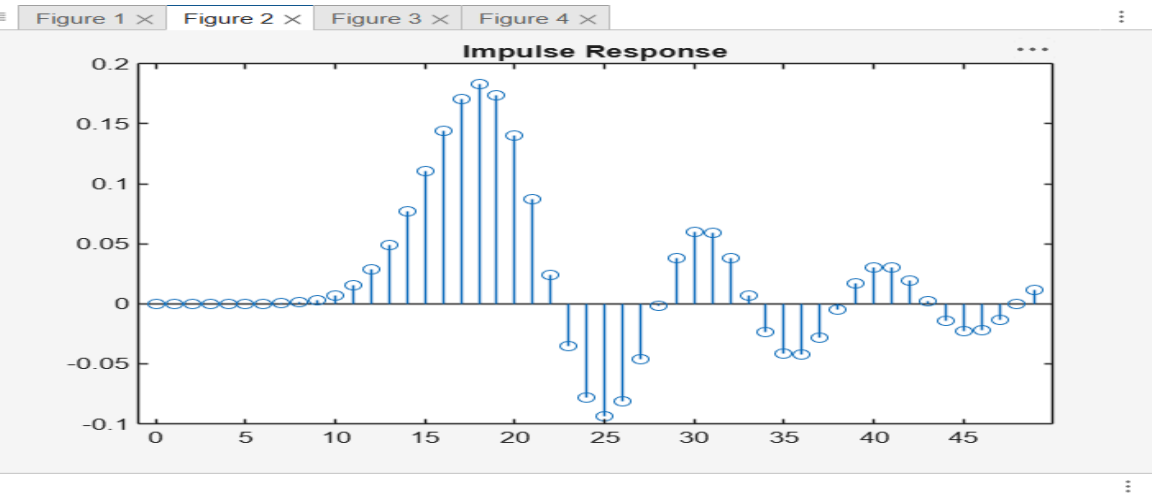
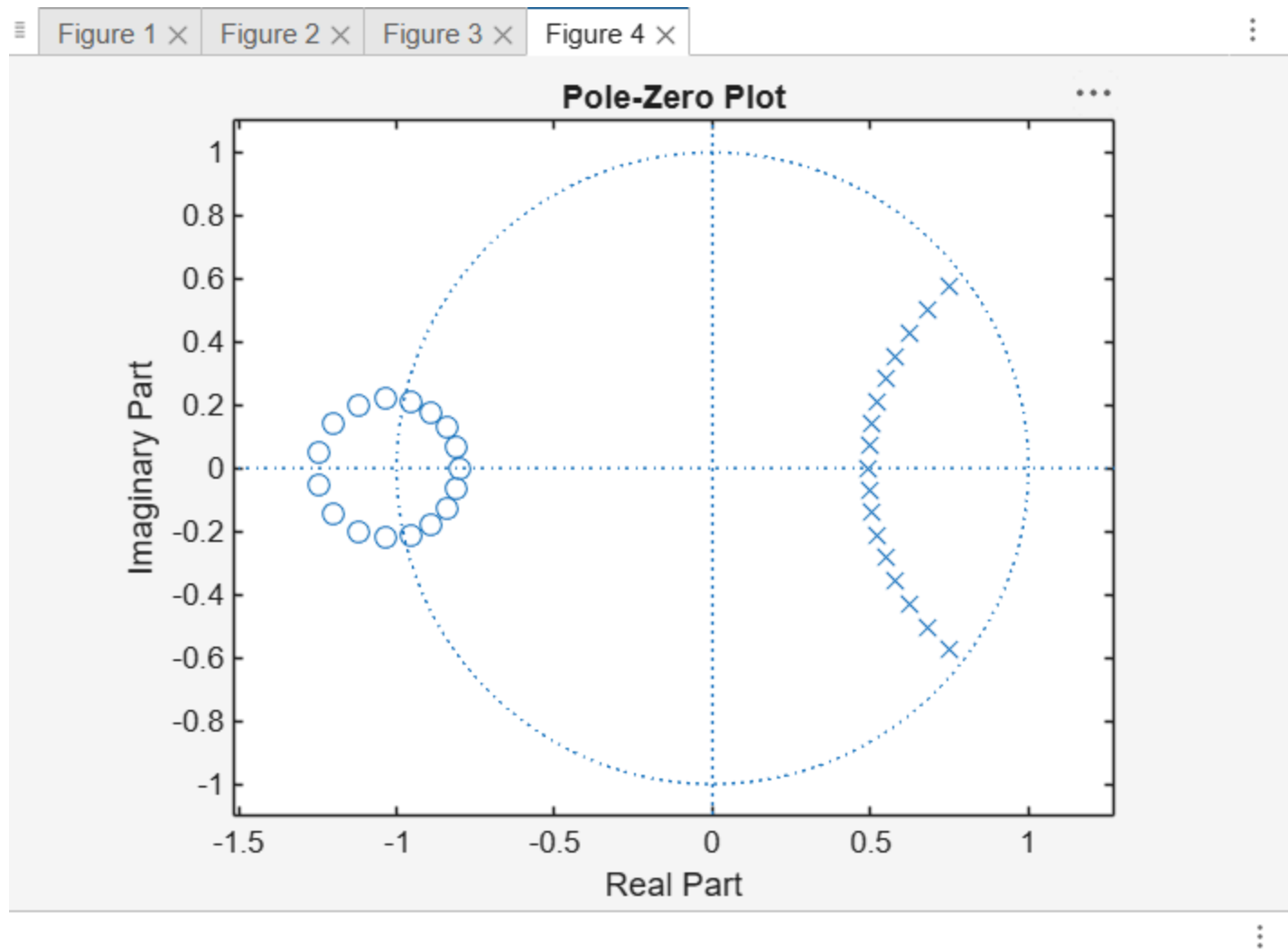


EXP 1





EXP 1.2

untitled13.m × untitled10.m × untitled12.m × +

ATLAB Drive/untitled12.m

```
1  clc;
2  clear;
3
4  Wp=0.4; Ws=0.55; Rp=1; As=40;
5
6  [N,Wn]=cheb1ord(Wp,Ws,Rp,As);
7  [b,a]=cheby1(N,Rp,Wn);
8
9  fprintf('Filter Order = %d\n',N);
10 fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);
11
12 if all(abs(roots(a))<1)
13     disp('System is STABLE');
14 else
15     disp('System is UNSTABLE');
16 end%
17 % ----- GRAPHS -----
18 figure; freqz(b,a); % Magnitude + Phase
19 figure; impz(b,a); % Impulse Response
20 figure; grpdelay(b,a); % Group Delay
21 figure; zplane(b,a); % Pole-Zero
```

Figure 1 × Figure 2 × Figure 3 × Figure 4 ×

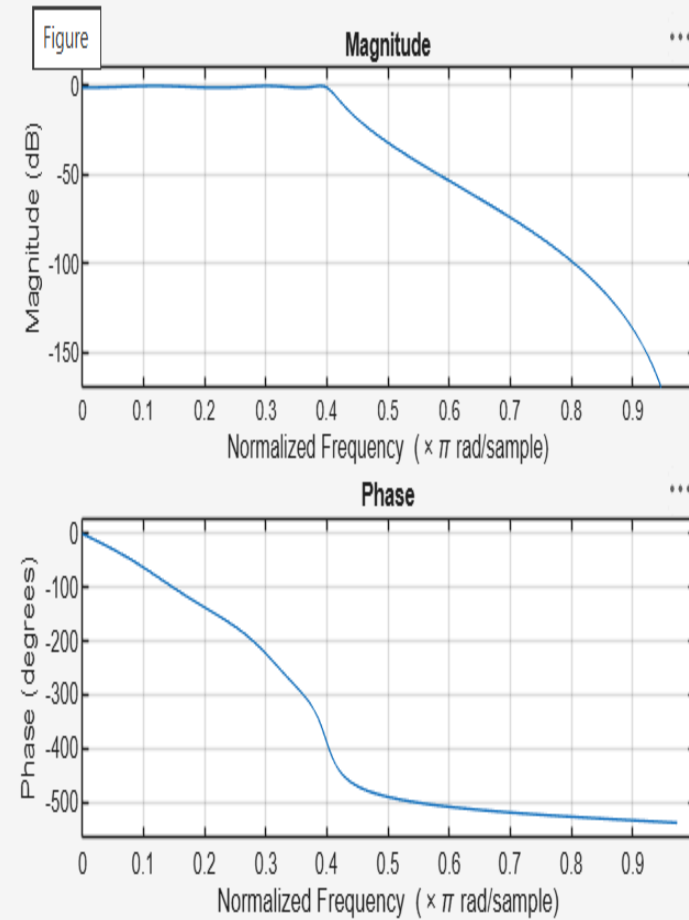
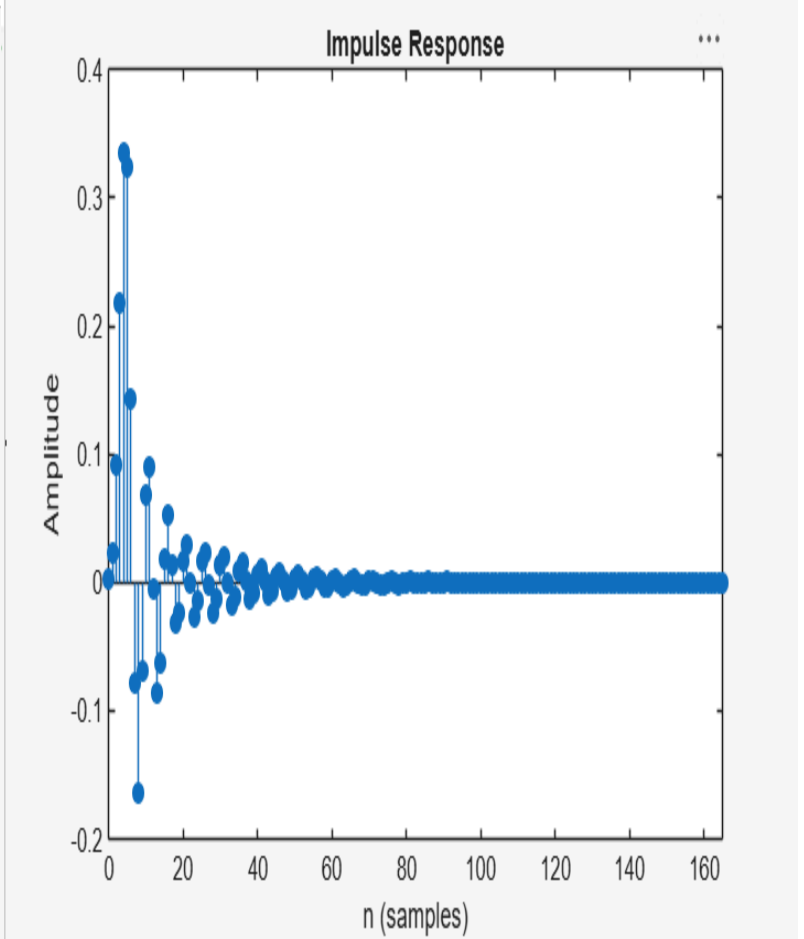
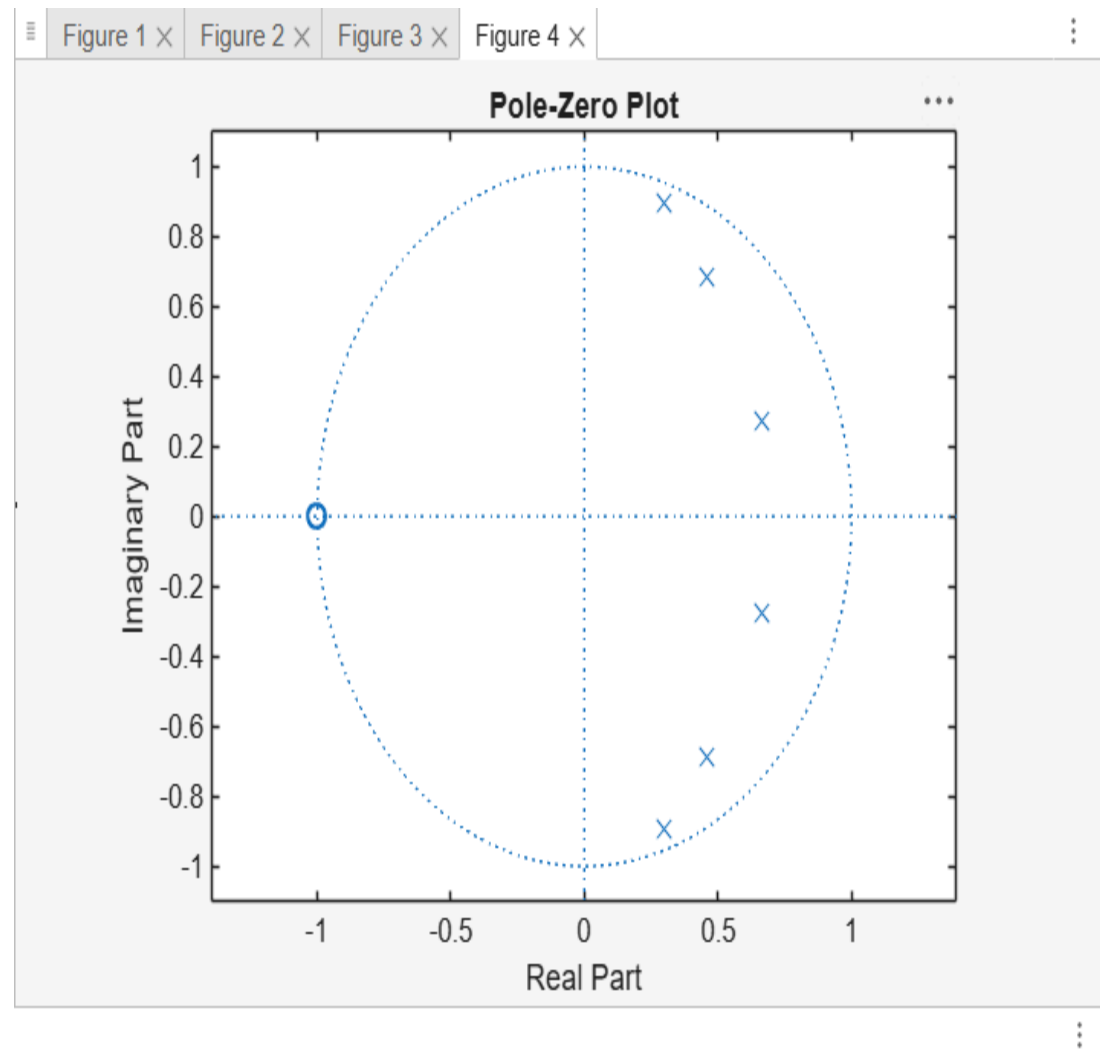
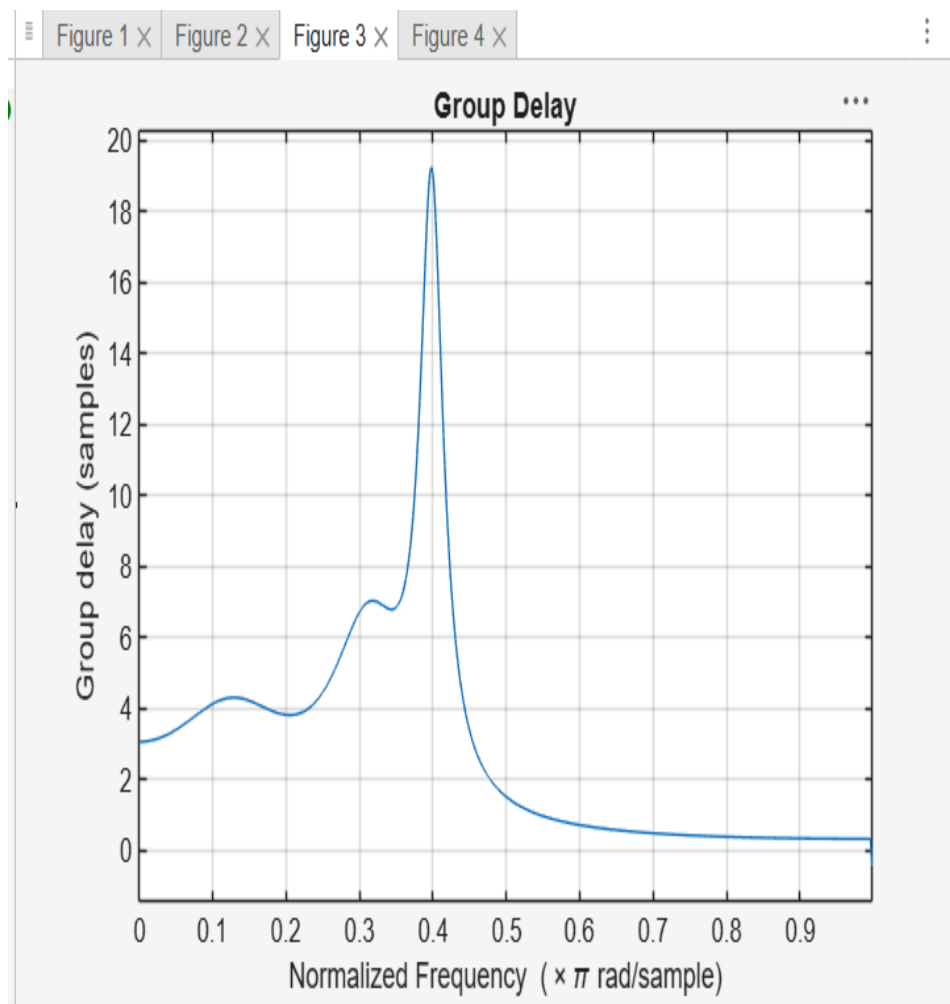


Figure 1 × Figure 2 × Figure 3 × Figure 4 ×





EXP 2

rive/untitled12.m

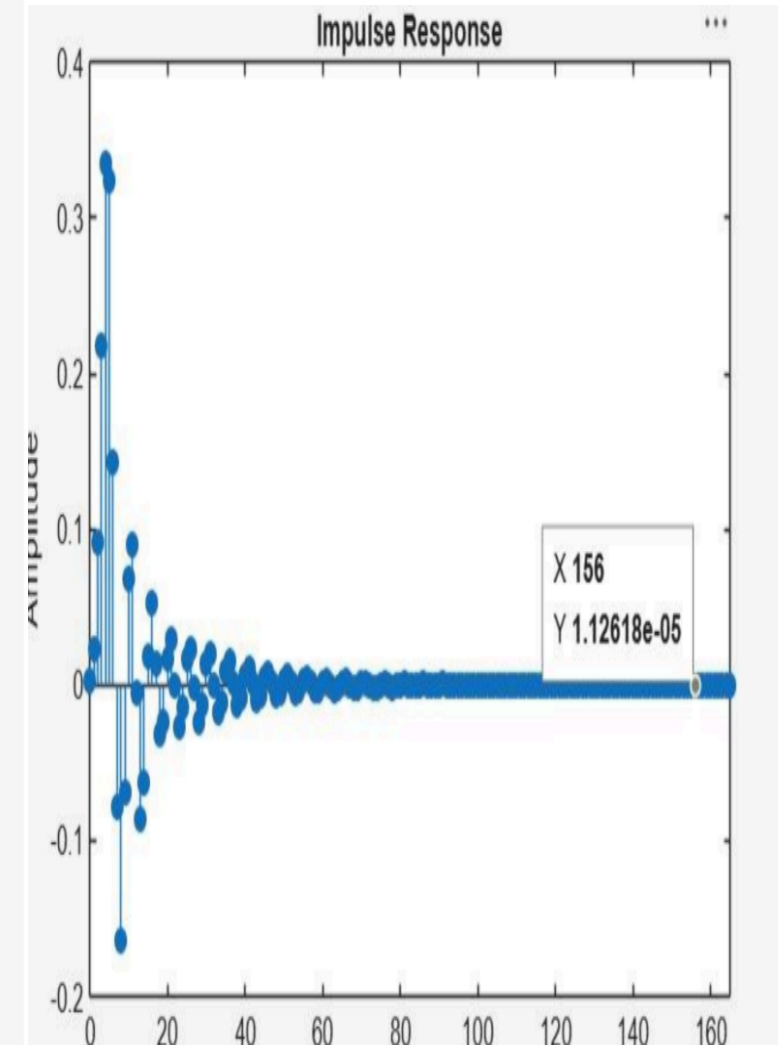
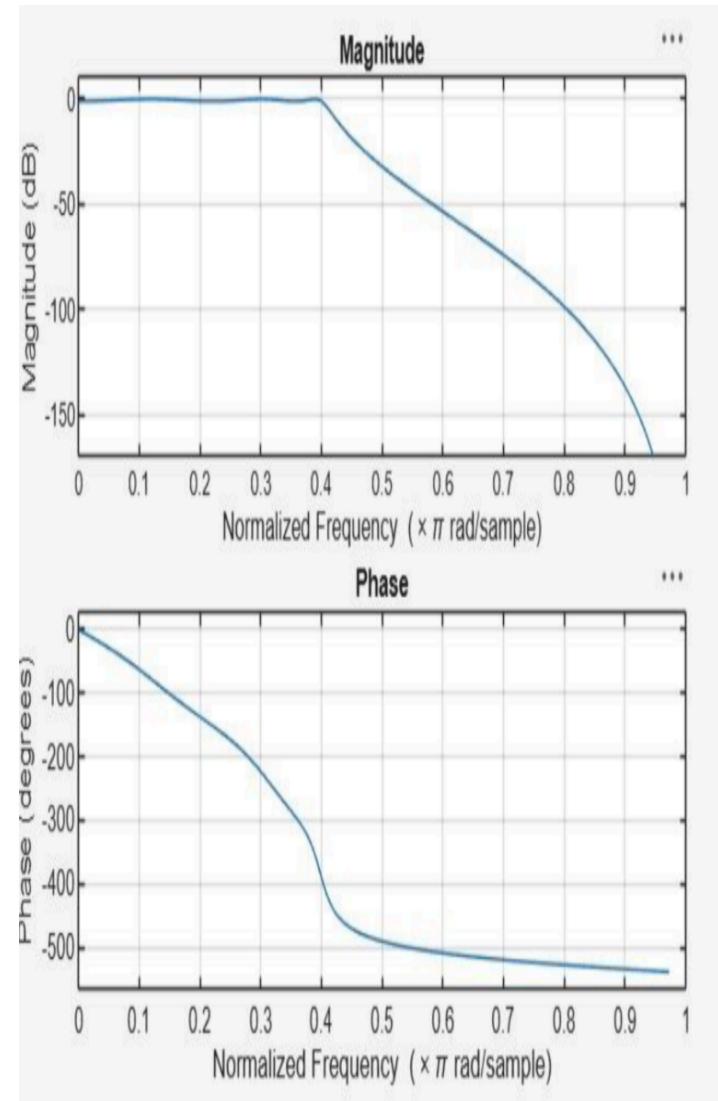
```
clc;
clear;

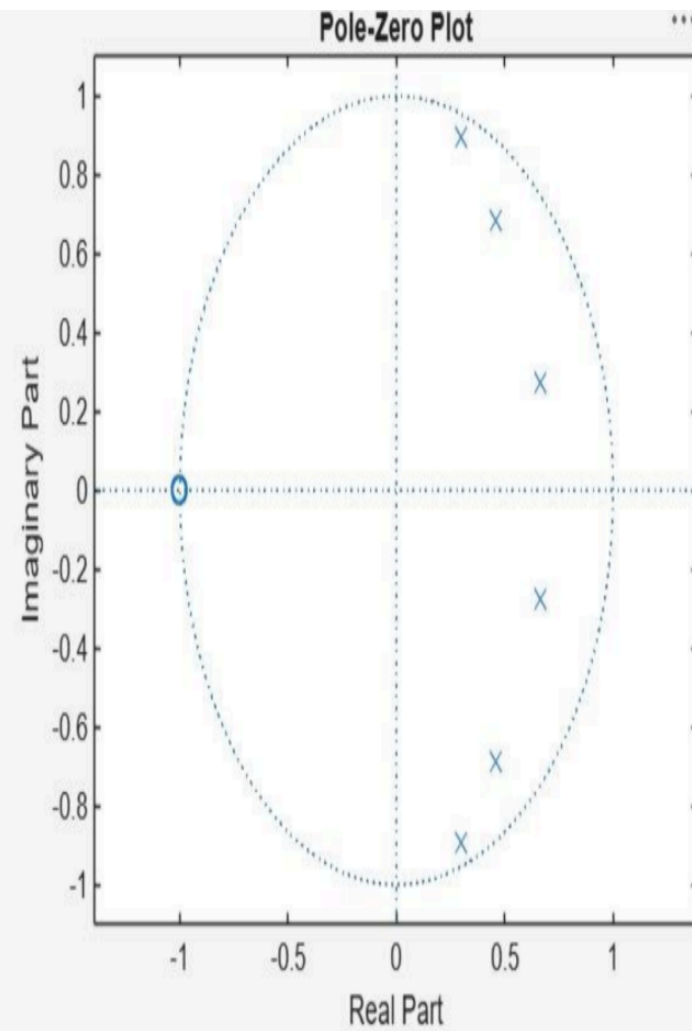
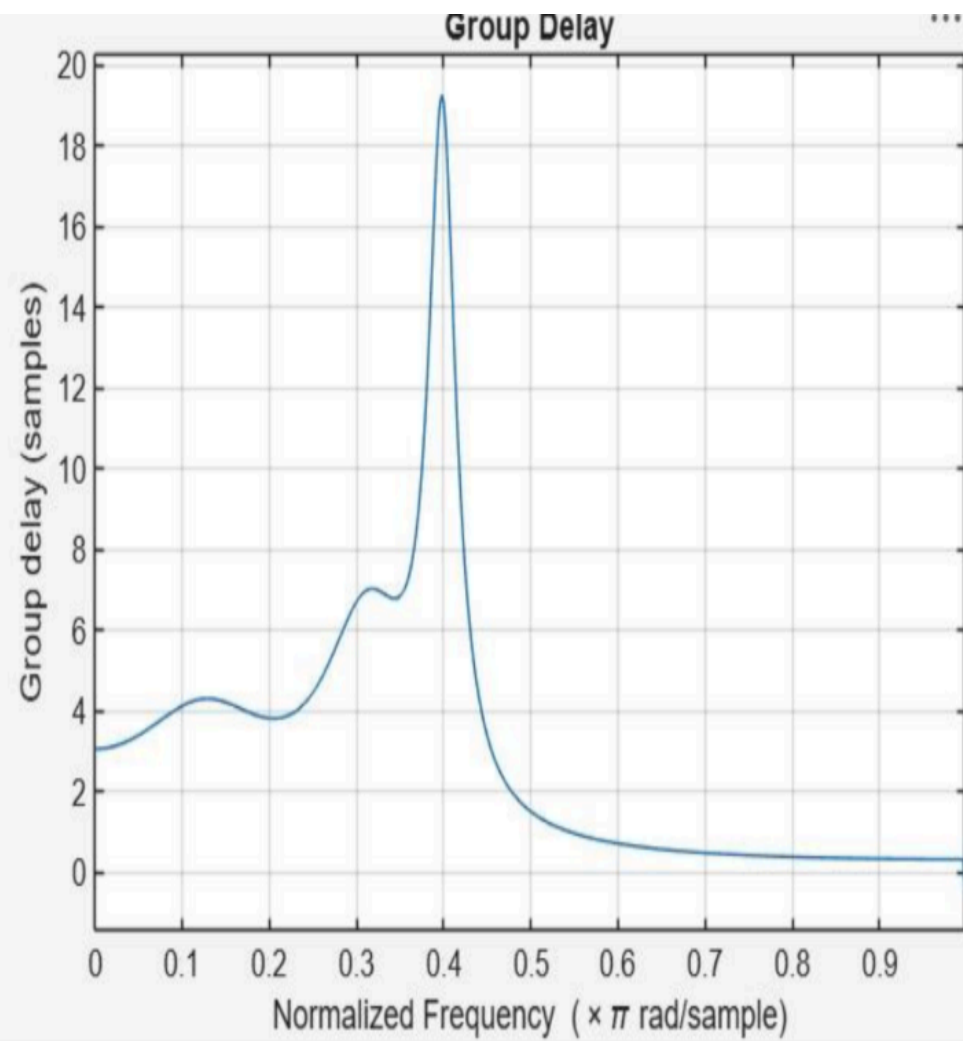
Wp=0.4; Ws=0.55; Rp=1; As=40;

[N,Wn]=cheb1ord(Wp,Ws,Rp,As);
[b,a]=cheby1(N,Rp,Wn);

fprintf('Filter Order = %d\n',N);
fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);

if all(abs(roots(a))<1)
    disp('System is STABLE');
else
    disp('System is UNSTABLE');
end%
% ----- GRAPHS -----
figure; freqz(b,a); % Magnitude + Phase
figure; impz(b,a); % Impulse Response
figure; grpdelay(b,a); % Group Delay
figure; zplane(b,a); % Pole-Zero
```





EXP 2.1

```
clc;
clear;
close all;

% Specifications
Rp = 1; % Passband Ripple (dB)
Rs = 40; % Stopband Attenuation (dB)
Wp = 0.5; % Passband Edge

Ws = 0.35; % Stopband Edge

% Find Filter Order
[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);

% Design High-Pass Filter
[b,a] = cheby1(n,Rp,Wn,'high');

% Frequency Response
figure;
freqz(b,a);
title('Frequency Response');

% Impulse Response
figure;
impz(b,a);
title('Impulse Response');

% Group Delay
figure;
grpdelay(b,a);
title('Group Delay');

% Pole-Zero Plot
```

